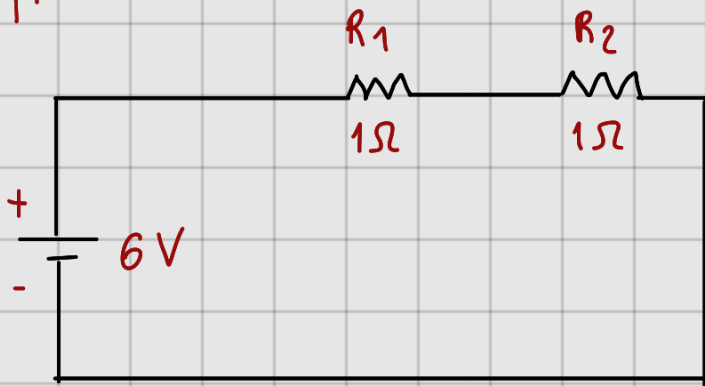


1.-



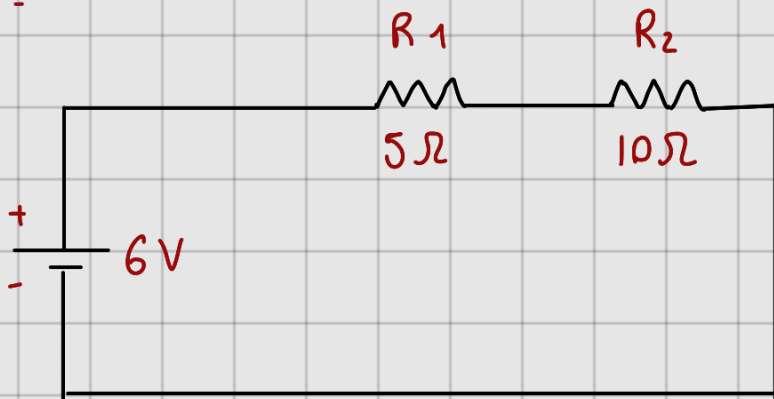
$$R_T = 2\Omega$$

$$I_T = 3A$$

$$R_T = R_1 + R_2 = 1\Omega + 1\Omega = 2\Omega$$

$$I_T = \frac{V}{R_T} = \frac{6V}{2\Omega} = 3A$$

2.-



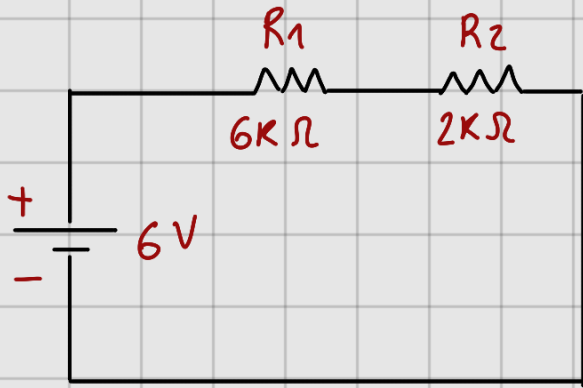
$$R_T = 15\Omega$$

$$I_T = 0.4A$$

$$R_T = R_1 + R_2 = 5\Omega + 10\Omega = 15\Omega$$

$$I_T = \frac{V}{R_T} = \frac{6V}{15\Omega} = 0.4A$$

3.-



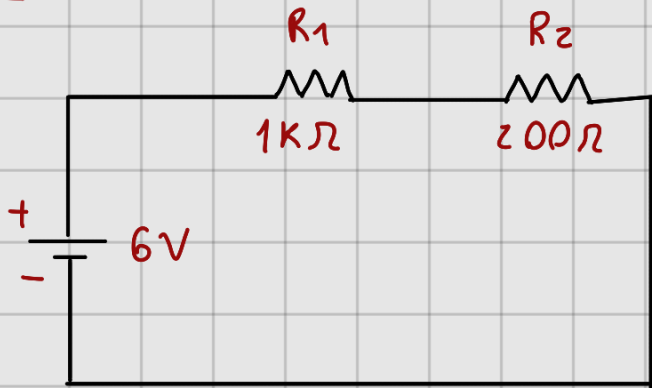
$$R_T = 8k\Omega$$

$$I_T = 750\mu A$$

$$R_T = 6k\Omega + 2k\Omega = 8k\Omega$$

$$I_T = \frac{V}{R_T} = \frac{6V}{8k\Omega} = 750\mu A$$

4.-



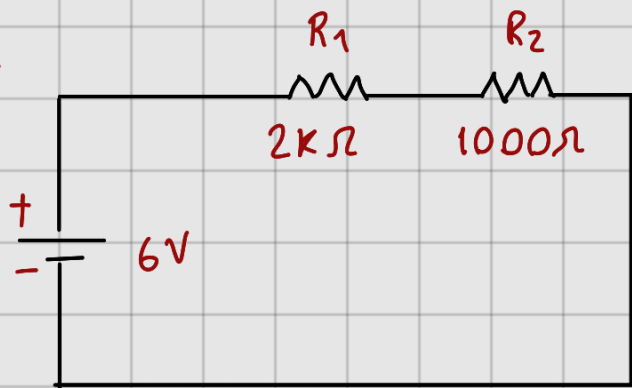
$$R_T = 1.2k\Omega$$

$$I_T = 5mA$$

$$R_T = 1k\Omega + 200\Omega = 1.2k\Omega$$

$$I_T = \frac{V}{R_T} = \frac{6V}{1.2k\Omega} = 5mA$$

5.-



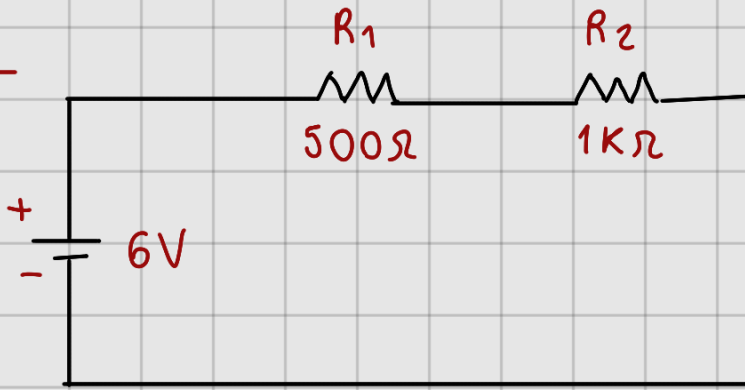
$$R_T = 3k\Omega$$

$$I_T = 2mA$$

$$R_T = 2k\Omega + 1k\Omega = 3k\Omega$$

$$I_T = \frac{V}{R_T} = \frac{6V}{3k\Omega} = 2mA$$

6.-



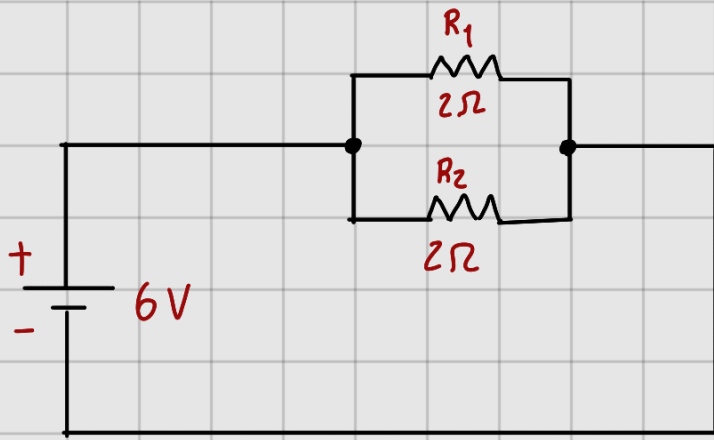
$$R_T = 1.5k\Omega$$

$$I_T = 4mA$$

$$R_T = 500\Omega + 1k\Omega = 1.5k\Omega$$

$$I_T = \frac{V}{R_T} = \frac{6V}{1.5k\Omega} = 4mA$$

7.-



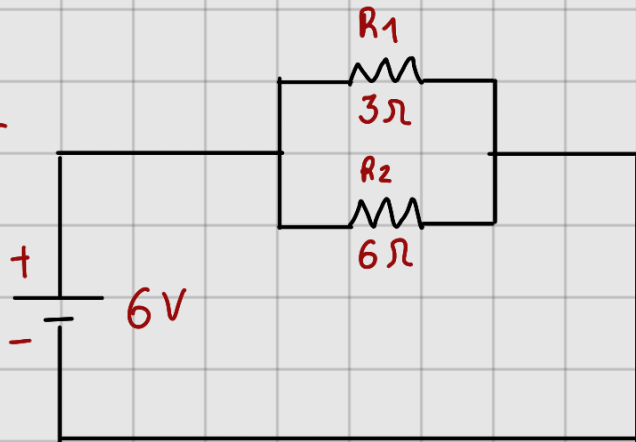
$$R_T = 1\Omega$$

$$I_T = 6A$$

$$R_T = \frac{1}{\frac{1}{2\Omega} + \frac{1}{2\Omega}} = \frac{1}{1\Omega} = 1\Omega$$

$$I_T = \frac{V}{R_T} = \frac{6V}{1\Omega} = 6A$$

8.-



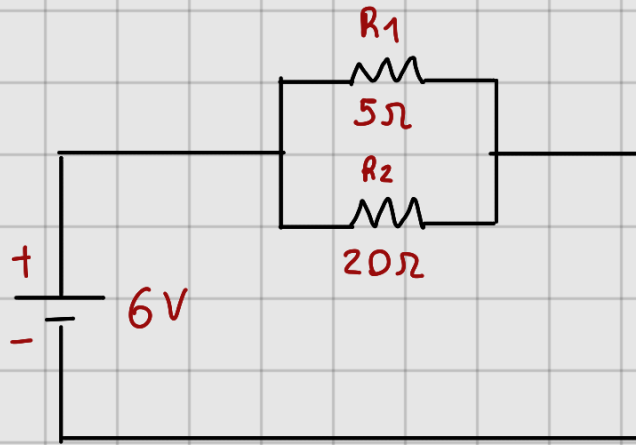
$$R_T = 2\Omega$$

$$I_T = 3A$$

$$R_T = \frac{1}{\frac{1}{3\Omega} + \frac{1}{6\Omega}} = \frac{1}{0.5\Omega} = 2\Omega$$

$$I_T = \frac{V}{R_T} = \frac{6V}{2\Omega} = 3A$$

9.-



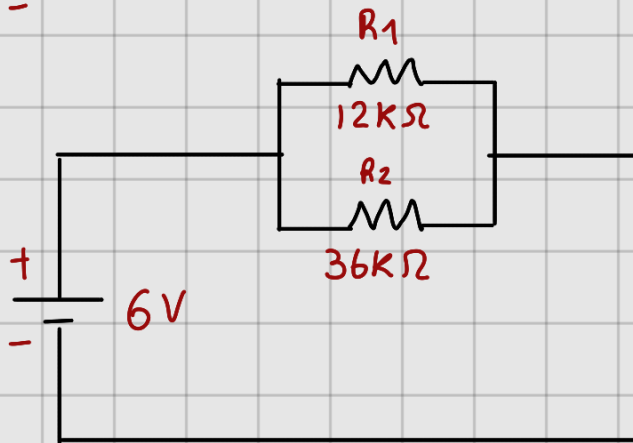
$$R_T = 4\Omega$$

$$I_T = 1.5 A$$

$$R_T = \frac{1}{\frac{1}{5\Omega} + \frac{1}{20\Omega}} = \frac{1}{0.25\Omega} = 4\Omega$$

$$I_T = \frac{V}{R_T} = \frac{6V}{4\Omega} = 1.5 A$$

10.-



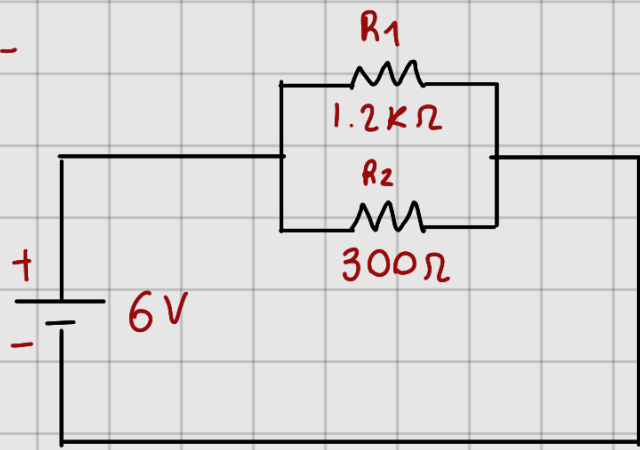
$$R_T = 9K\Omega$$

$$I_T = 666\mu A$$

$$R_T = \frac{1}{\frac{1}{12000\Omega} + \frac{1}{36000\Omega}} = \frac{1}{111\mu\Omega} = 9K\Omega$$

$$I_T = \frac{V}{R_T} = \frac{6V}{9K\Omega} = 666\mu A$$

11..



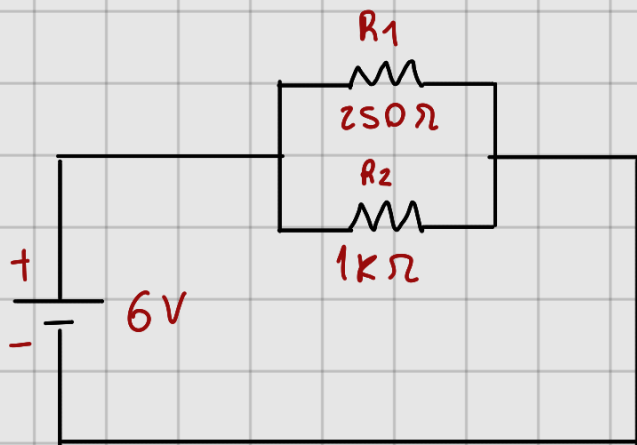
$$R_T = 240\Omega$$

$$I_T = 25mA$$

$$R_T = \frac{1}{\frac{1}{1200} + \frac{1}{300}} = \frac{1}{4.16m\Omega} = 240\Omega$$

$$I_T = \frac{V}{R_T} = \frac{6V}{240\Omega} = 25mA$$

12..



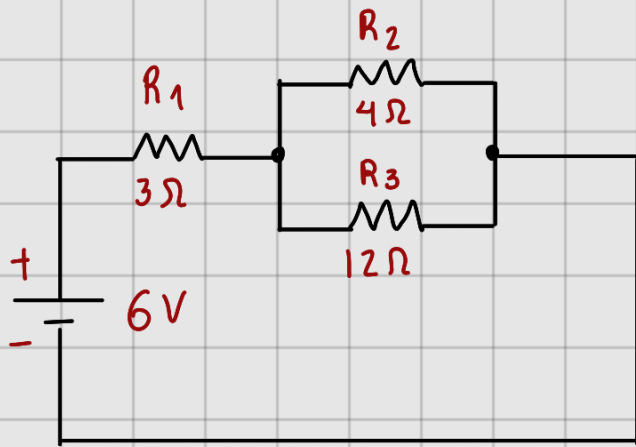
$$R_T = 200\Omega$$

$$I_T = 30mA$$

$$R_T = \frac{1}{\frac{1}{1000} + \frac{1}{250}} = \frac{1}{5m\Omega} = 200\Omega$$

$$I_T = \frac{V}{R_T} = \frac{6V}{200\Omega} = 30mA$$

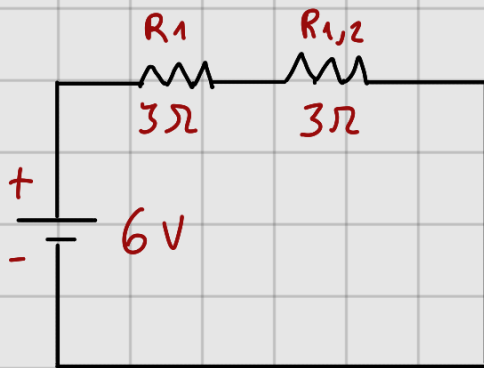
13..



$$R_T = 6\Omega$$

$$I_T = 1A$$

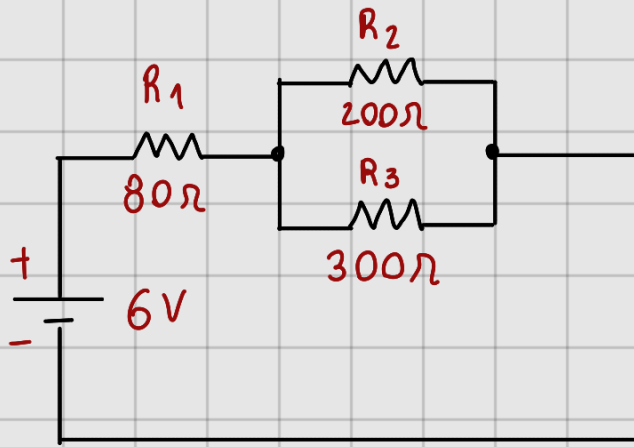
$$R_{2,3} = \frac{1}{\frac{1}{4\Omega} + \frac{1}{12\Omega}} = \frac{1}{0.33\overline{3}\Omega} = 3\Omega$$



$$R_T = 3\Omega + 3\Omega = 6\Omega$$

$$I_T = \frac{V}{R_T} = \frac{6V}{6\Omega} = 1A$$

14.-



$$R_T = 120\Omega$$

$$I_T = 30\text{mA}$$

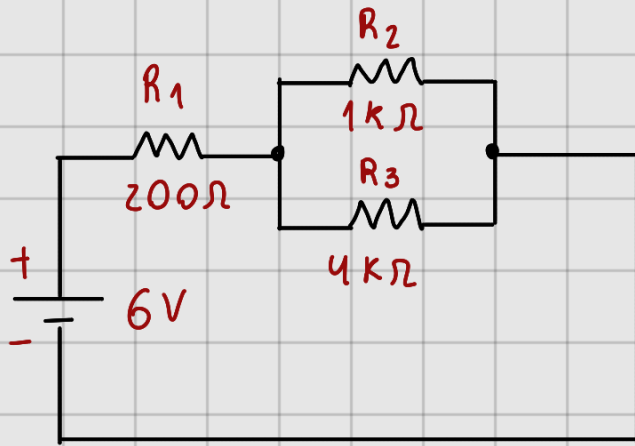
$$R_{2,3} = \frac{1}{\frac{1}{200\Omega} + \frac{1}{300\Omega}} = \frac{1}{8.3\text{m}\Omega} = 120\Omega$$



$$R_T = 80\Omega + 120\Omega = 200\Omega$$

$$I_T = \frac{V}{R_T} = \frac{6\text{V}}{200\Omega} = 30\text{mA}$$

15.-



$$R_T = 1k\Omega$$

$$I_T = 6mA$$

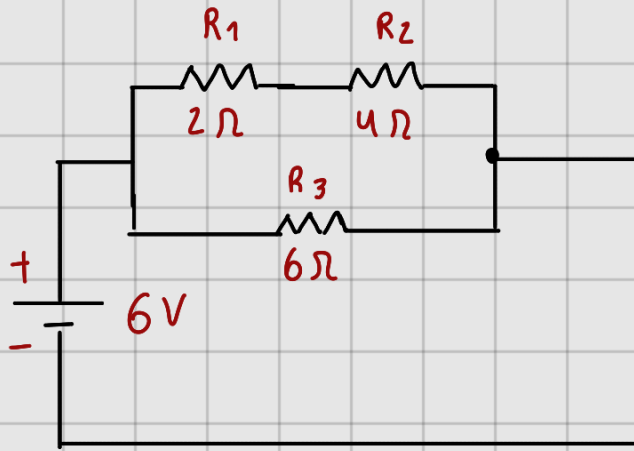
$$R_{2,3} = \frac{1}{\frac{1}{4000\Omega} + \frac{1}{1000\Omega}} = \frac{1}{1.25m\Omega} = 800\Omega$$



$$R_T = 200\Omega + 800\Omega = 1k\Omega$$

$$I_T = \frac{V}{R_T} = \frac{6V}{1k\Omega} = 6mA$$

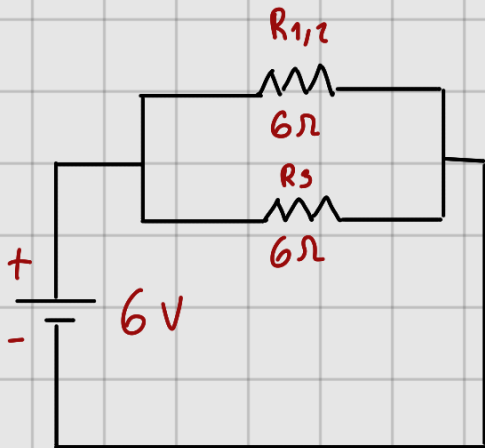
16..



$$R_T = 3\Omega$$

$$I_T = 2A$$

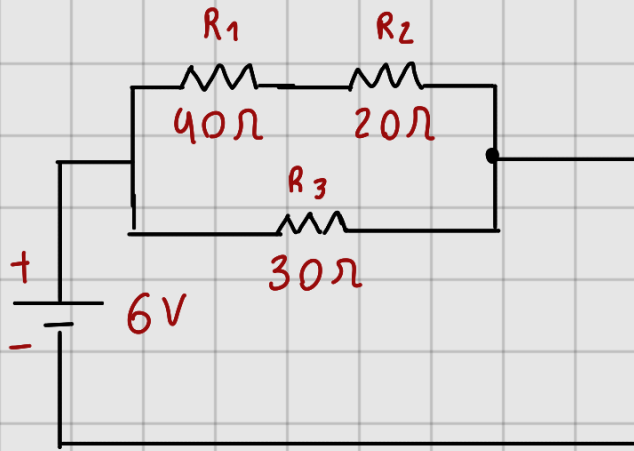
$$R_{1,2} = 2\Omega + 4\Omega = 6\Omega$$



$$R_T = \frac{1}{\frac{1}{6\Omega} + \frac{1}{6\Omega}} = \frac{1}{0.33\Omega} = 3\Omega$$

$$I_T = \frac{V}{R_T} = \frac{6V}{3\Omega} = 2A$$

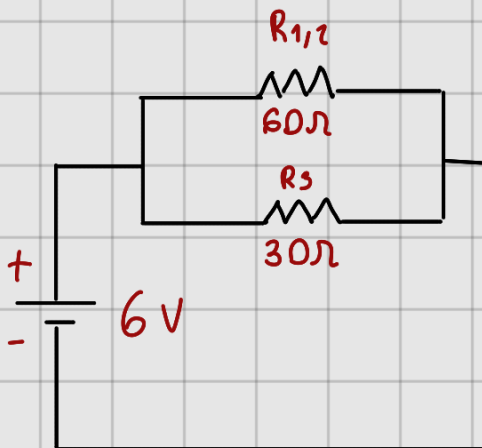
17..



$$R_T = 20\Omega$$

$$I_T = 0.3A$$

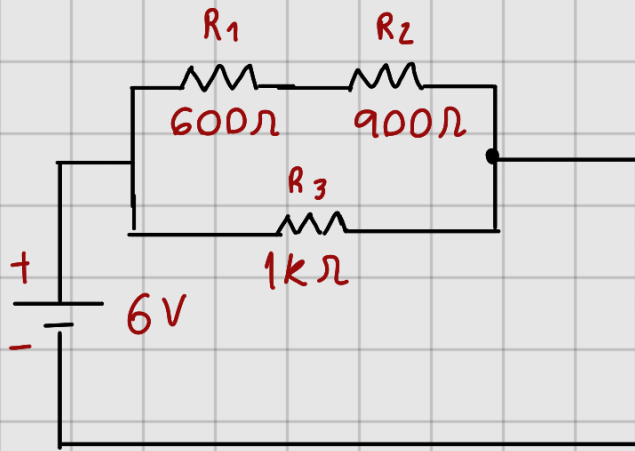
$$R_{1,2} = 40\Omega + 20\Omega = 60\Omega$$



$$R_T = \frac{1}{\frac{1}{60\Omega} + \frac{1}{30\Omega}} = \frac{1}{50m\Omega} = 20\Omega$$

$$I_T = \frac{V}{R_T} = \frac{6V}{20\Omega} = 0.3A$$

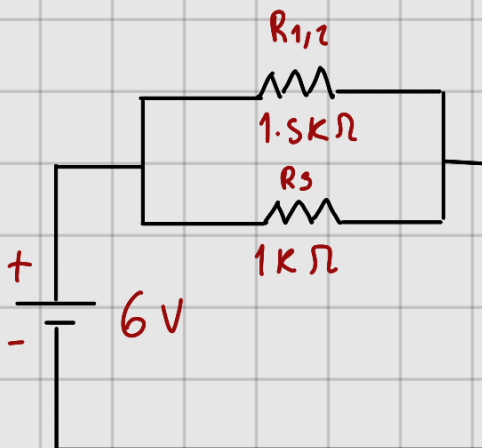
18..



$$R_T = 600\Omega$$

$$I_T = 10mA$$

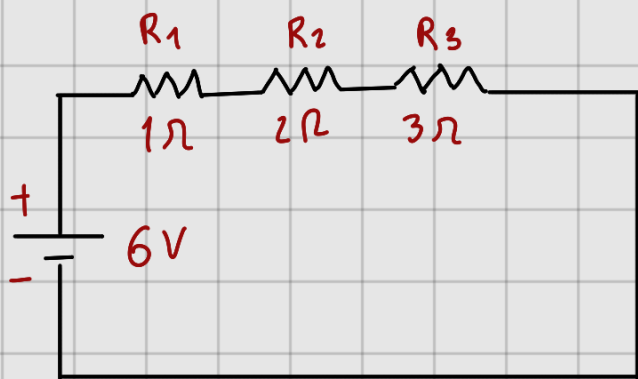
$$R_{1,2} = 600\Omega + 900\Omega = 1.5k\Omega$$



$$R_T = \frac{1}{\frac{1}{1000} + \frac{1}{1500}} = \frac{1}{1.6m\Omega} = 600\Omega$$

$$I_T = \frac{V}{R_T} = \frac{6V}{600\Omega} = 10mA$$

19..



$$R_T = 6\Omega$$

$$I_T = 1A$$

$$R_T = 1\Omega + 2\Omega + 3\Omega = 6\Omega$$

$$I = \frac{V}{R_T} = \frac{6V}{6\Omega} = 1A$$

20..



$$R_T = 1.2K\Omega$$

$$I_T = 5mA$$

$$R_T = 200\Omega + 600\Omega + 400\Omega = 1.2K\Omega$$

$$I = \frac{V}{R_T} = \frac{6V}{1.2K\Omega} = 5mA$$

21..



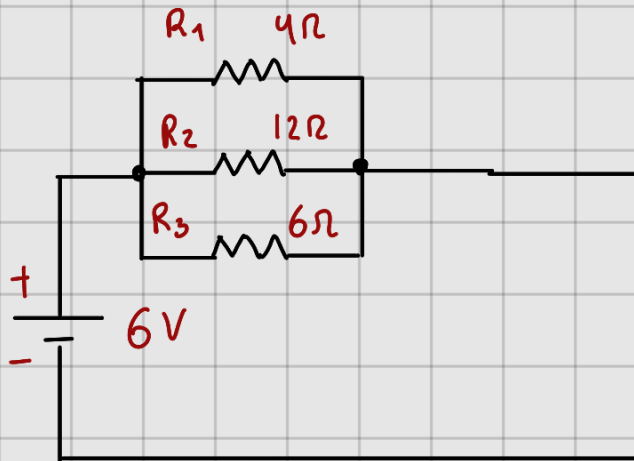
$$R_T = 3k\Omega$$

$$I_T = 2mA$$

$$R_T = 500\Omega + 500\Omega + 2000\Omega = 3k\Omega$$

$$I = \frac{V}{R_T} = \frac{6V}{3k\Omega} = 2mA$$

22..



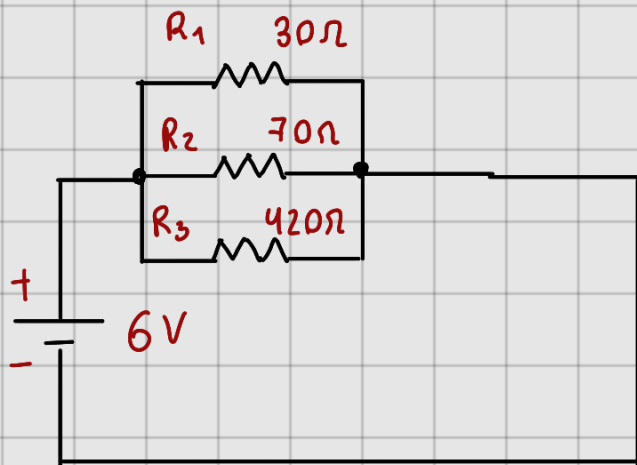
$$R_T = 2\Omega$$

$$I_T = 3A$$

$$R_T = \frac{1}{\frac{1}{4\Omega} + \frac{1}{12\Omega} + \frac{1}{6\Omega}} = 2\Omega$$

$$I_T = \frac{V}{R_T} = \frac{6V}{2\Omega} = 3A$$

23.-



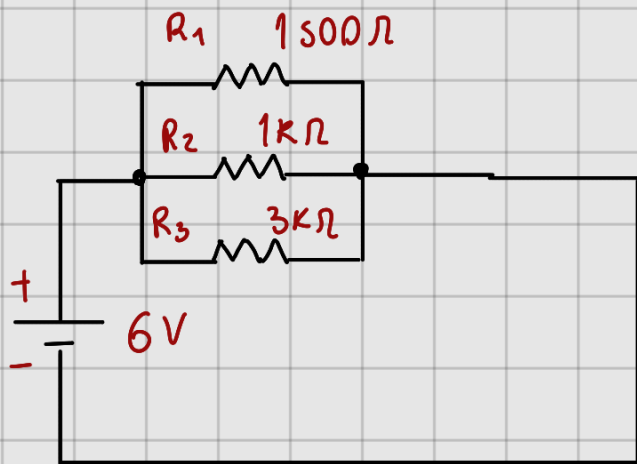
$$R_T = 20\Omega$$

$$I_T = 0.3A$$

$$R_T = \frac{1}{\frac{1}{30\Omega} + \frac{1}{70\Omega} + \frac{1}{420\Omega}} = 20\Omega$$

$$I_T = \frac{V}{R_T} = \frac{6V}{20\Omega} = 0.3A$$

24.-



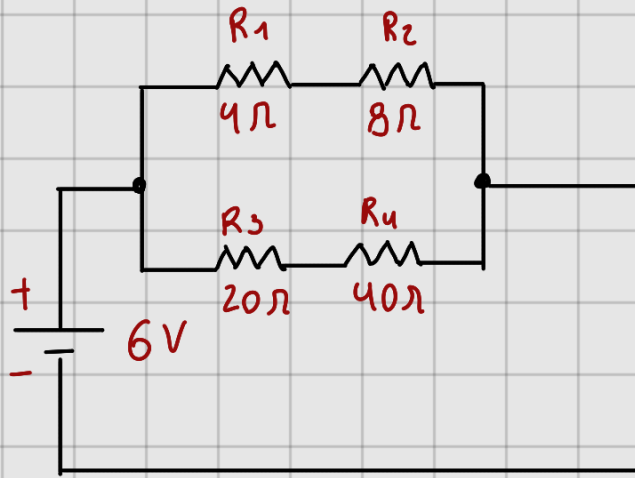
$$R_T = 500\Omega$$

$$I_T = 12mA$$

$$R_T = \frac{1}{\frac{1}{1500\Omega} + \frac{1}{1000\Omega} + \frac{1}{3000\Omega}} = 500\Omega$$

$$I_T = \frac{V}{R_T} = \frac{6V}{500\Omega} = 12mA$$

25..

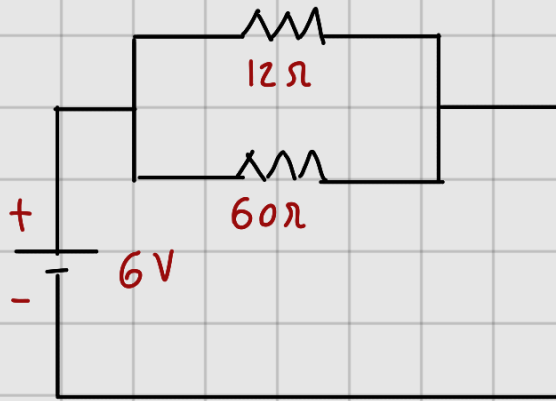


$$R_T = 10\Omega$$

$$I_T = 0.6A$$

$$R_{1,2} = 4\Omega + 8\Omega = 12\Omega$$

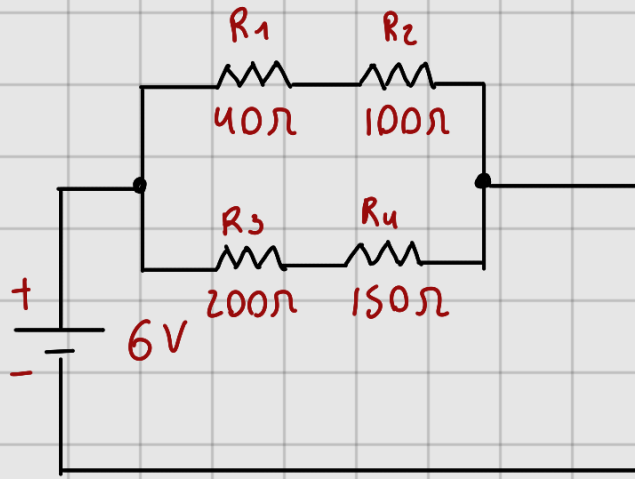
$$R_{3,4} = 20\Omega + 40\Omega = 60\Omega$$



$$R_T = \frac{1}{\frac{1}{12\Omega} + \frac{1}{60\Omega}} = 10\Omega$$

$$I = \frac{V}{R_T} = \frac{6V}{10\Omega} = 0.6A$$

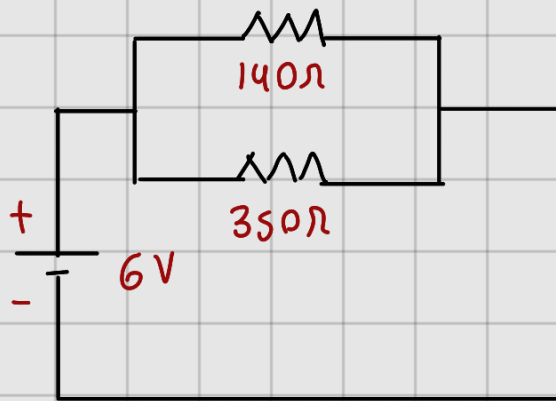
26. -



$$R_T = 100\Omega$$

$$I_T = 60\text{mA}$$

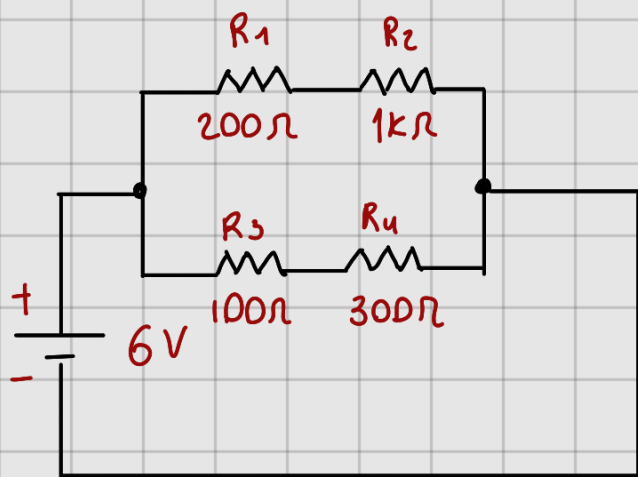
$$R_{1,2} = 40\Omega + 100\Omega = 140\Omega \quad R_{3,4} = 200\Omega + 150\Omega = 350\Omega$$



$$R_T = \frac{1}{\frac{1}{140\Omega} + \frac{1}{350\Omega}} = 100\Omega$$

$$I = \frac{V}{R_T} = \frac{6\text{V}}{100\Omega} = 60\text{mA}$$

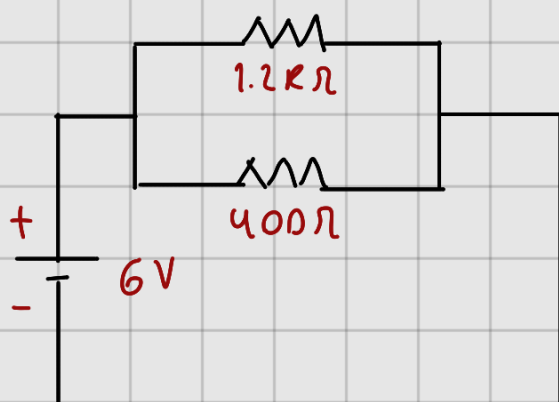
27.-



$$R_T = 300\Omega$$

$$I_T = 20\text{mA}$$

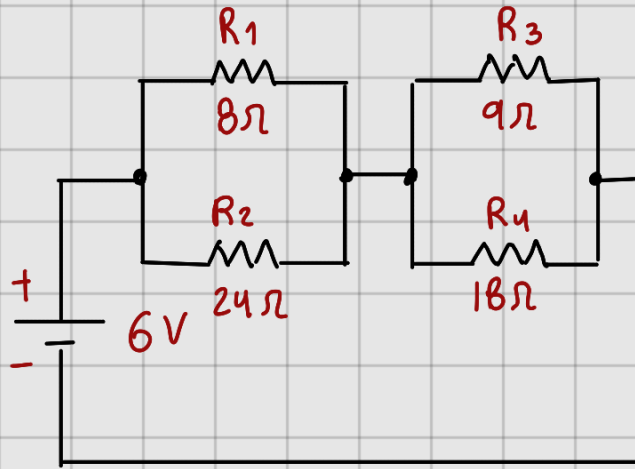
$$R_{1,2} = 200\Omega + 1000\Omega = 1200\Omega \quad R_{3,4} = 100\Omega + 300\Omega = 400\Omega$$



$$R_T = \frac{1}{\frac{1}{1200\Omega} + \frac{1}{400\Omega}} = 300\Omega$$

$$I = \frac{V}{R_T} = \frac{6\text{V}}{300\Omega} = 20\text{mA}$$

28.-

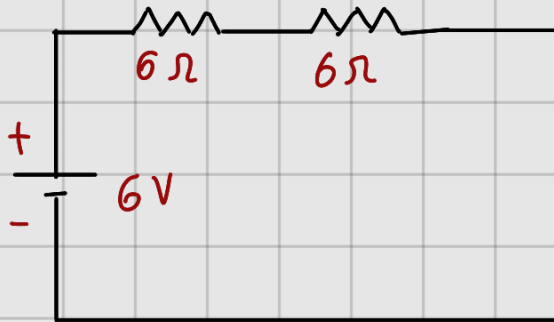


$$R_T = 12\Omega$$

$$I_T = 0.5A$$

$$R_{1,2} = \frac{1}{\frac{1}{8\Omega} + \frac{1}{24\Omega}} = 6\Omega$$

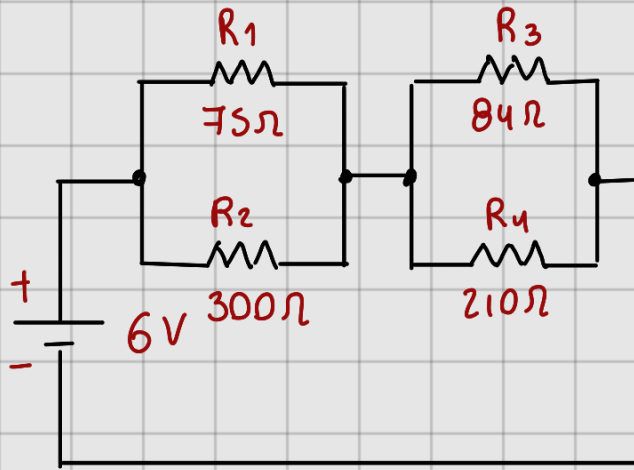
$$R_{3,4} = \frac{1}{\frac{1}{9\Omega} + \frac{1}{18\Omega}} = 6\Omega$$



$$R_T = 6\Omega + 6\Omega = 12\Omega$$

$$I_T = \frac{V}{R_T} = \frac{6V}{12\Omega} = 0.5A$$

29.-

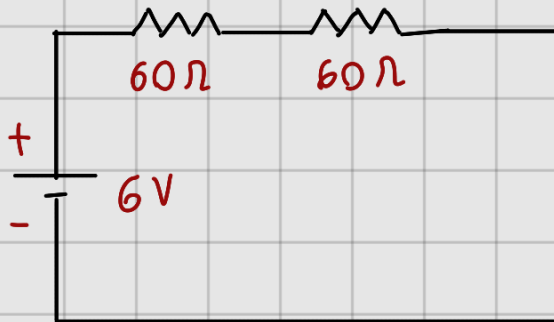


$$R_T = 120\Omega$$

$$I_T = 50\text{mA}$$

$$R_{1,2} = \frac{1}{\frac{1}{75\Omega} + \frac{1}{300\Omega}} = 60\Omega$$

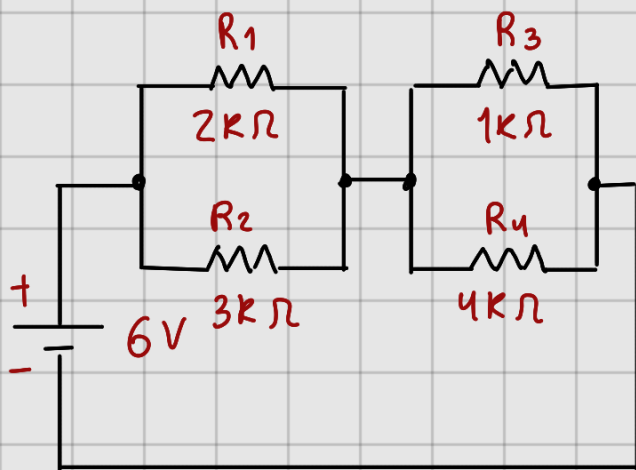
$$R_{3,4} = \frac{1}{\frac{1}{84\Omega} + \frac{1}{210\Omega}} = 60\Omega$$



$$R_T = 60\Omega + 60\Omega = 120\Omega$$

$$I_T = \frac{V}{R_T} = \frac{6\text{V}}{120\Omega} = 50\text{mA}$$

30.-



$$R_T = 2\text{k}\Omega$$

$$I_T = 3\text{mA}$$

$$R_{1,2} = \frac{1}{\frac{1}{2000\Omega} + \frac{1}{3000\Omega}} = 1.2\text{k}\Omega$$

$$R_{3,4} = \frac{1}{\frac{1}{1000\Omega} + \frac{1}{4000\Omega}} = 800\Omega$$



$$R_T = 1200\Omega + 800\Omega = 2\text{k}\Omega$$

$$I_T = \frac{V}{R_T} = \frac{6\text{V}}{2\text{k}\Omega} = 3\text{mA}$$